

Helen Peng (she/her)

helen.peng04@gmail.com • [linkedin.com/in/helenpeng04](https://www.linkedin.com/in/helenpeng04) • helenpeng04.github.io

EDUCATION

Columbia University Mailman School of Public Health

Master of Science in Biostatistics (Pharmaceutical Statistics Track)

New York, NY
Expected May 2028

Carnegie Mellon University

Bachelor of Science, Statistics & Data Science

Pittsburgh, PA
December 2025

Coursework: Statistical Methods in Epidemiology, Statistical Machine Learning, Casual Inference, Modern Regression, Meta-Analysis, Statistical Inference, Probability Theory

Honors

Dean's List, High Honors: Fall 2022, Spring 2025, Fall 2025

Dean's List: Spring 2023, Spring 2024

RESEARCH EXPERIENCE

UnitedHealth Group Bridges to Healthcare Technology Research Program, Carnegie Mellon University

Pittsburgh, PA

Research Intern

Summer 2025

- Performed exploratory data analysis and regression modeling to identify predictors of obesity in a collaborative healthcare analytics project; communicated findings to industry stakeholders and peers.
- Applied k-means clustering and EDA to analyze COVID-19 case and mortality trends across Pennsylvania counties to support population-level pattern discovery.

Optimized Algorithms and Knowledge (OAK) Lab, Carnegie Mellon University

Pittsburgh, PA

Research Assistant

Spring 2023 – Fall 2025

- Applied text mining and clustering methods to structure and analyze 10,000+ qualitative participant responses for behavioral research.
- Implemented multilevel (hierarchical) modeling on data from 150+ participants to evaluate intervention effects on learning outcomes under different experimental conditions.
- Designed and analyzed Qualtrics-based behavioral studies, integrating survey methodology with statistical evaluation of learning and motivation constructs.
- Communicated reproducible statistical findings through weekly progress updates and formal presentations to research group members.
- Processed, anonymized, and managed 200+ assessment datasets ensuring compliance with research data privacy standards and reproducibility requirements.

TEACHING EXPERIENCE

Research Methods in Cognitive Psychology, Carnegie Mellon University

Pittsburgh, PA

Teaching Assistant

Spring 2025

- Supported psychology students with limited programming experience in R through debugging, statistical analysis, and interpretation of results; graded weekly assignments and provided feedback on applied research methods and experimental design.

PROFESSIONAL EXPERIENCE

Zhong Ou Asset Management Intl

Hong Kong

Financial Research Intern

Summer 2024

- Built Excel-based visualizations to compare investment performance using pivot tables, formulas, and data formatting techniques; supported preparation of monthly reports summarizing macroeconomic indicators and competitor trends in China.

VOLUNTEER AND LEADERSHIP EXPERIENCE

Students Using Data for Social Good, Carnegie Mellon University

Pittsburgh, PA

Data Analyst

Spring 2024 – Spring 2025

- Applied statistical methods including Fisher's exact test, Kruskal-Wallis test, and survival analysis to healthcare data from 600+ individuals with developmental disabilities to identify patterns in medication errors and associated risk factors.

- Developed data visualizations and analytical reports translating findings into actionable insights for nonprofit stakeholders to support service delivery and resource allocation decisions.

Cognitive Science Student Advisory, Carnegie Mellon University

Pittsburgh, PA

Executive Board Officer

Fall 2023 – Fall 2025

- Organized and promoted 15+ academic and social events connecting undergraduate students with graduate students and faculty in cognitive science and related fields.
- Initiated and launched a recurring end-of-semester student engagement event (“Boba Finals Pickup”) to support community building and student well-being during finals periods.
- Led onboarding of 5 new board members and mentored students in statistics and data analytics coursework and research exploration.

PROJECTS

Developmental trajectories of racial attitudes in young children

Fall 2025

- Analyzed experimental behavioral data on children’s racial attitudes using exploratory data analysis, logistic regression, and multinomial regression in R.
- Collaborated with a teammate to communicate findings to advisor and client through iterative presentations and a full research deliverable, including poster, written report, presentation, and recorded summary.
- Awarded First Place, Statistics Poster Competition, Meeting of the Minds Undergraduate Research Symposium, Carnegie Mellon University (Spring 2026).

Mental Health Outcomes of COVID-19 Graduates

Spring 2025

- Analyzed survey data from 100+ participants to evaluate differences in mental health outcomes across graduating cohorts.
- Developed and validated predictive models using PCA, logistic regression, and bootstrap methods in R.

Gender Stereotype Threat Interventions and Academic Performance

Spring 2024

- Conducted meta-analysis of 117 studies examining effects of stereotype threat on academic performance, extracting and synthesizing effect sizes from peer-reviewed literature.
- Performed statistical analysis in R to evaluate aggregated findings across studies.

Gender Stereotype Threat and Memory Performance in a Heading-Recall Task

Spring 2024

- Designed and executed a behavioral experiment (35 participants, 1,400+ trials) using the Gorilla experiment platform.
- Conducted data cleaning in Excel and performed ANOVA analysis in Jamovi; presented findings at departmental research session.
- Presented findings at Carnegie Mellon University Psychology Department undergraduate research symposium.

SKILL

Programming & Data Analysis

- R: tidyverse, data.table, caret, survival, statistical modeling, markdown reporting
- Python: Basic knowledge of pandas and numpy
- SQL: PostgreSQL querying, relational database management

Research Tools & Experimental Design

- Qualtrics, Gorilla: Survey and experimental protocol design, randomization, data collection
- LaTeX: Scientific writing and formatting
- GitHub: Version control and collaborative coding
- Excel: Pivot tables, advanced formulas, conditional formatting, charts
- Jupyter Notebook: Integrated R/Python for reproducible workflows
- Quarto: Markdown reporting, presentations

Languages

- English (Native), Mandarin Chinese (Heritage Proficiency), Spanish (Limited Proficiency)